

BAYANBULAG, Mongolia

WMO: 442750

Lat: 46.8122N

Lon: 98.0869E

Elev: 7398

StdP: 11.17

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.3	-28.0	-36.7	0.9	-30.9	-32.9	1.1	-26.4	25.9	3.0	22.0	4.1	1.0	50	0.787

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	75.7	57.6	72.7	55.8	69.9	54.4	60.7	72.8	58.3	69.9	56.2	67.0	9.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.8	87.5	68.5	53.4	80.1	65.5	51.1	73.6	63.1	30.8	73.0	28.8	70.2	27.2	67.1	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	26.1	21.7	19.3	DB	-36.8	81.5	4.7	3.1	-40.2	83.8	-42.9	85.6	-45.5	87.4	-49.0	89.7
(5)				WB	-36.6	64.0	4.7	5.2	-40.0	67.8	-42.7	70.9	-45.4	73.8	-48.8	77.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	-7.2	-0.8	13.1	30.9	42.0	53.2	57.5	53.1	40.8	25.9	9.5	-2.6	(6)
(7)		DBStd	23.90	8.66	9.84	10.43	8.85	7.78	5.74	5.29	5.29	7.14	8.53	10.19	8.71	(7)
(8)		HDD50	9117	1774	1423	1143	574	263	32	5	29	283	746	1214	1631	(8)
(9)		HDD65	14088	2239	1843	1608	1023	713	356	240	369	726	1211	1664	2096	(9)
(10)		CDD50	513	0	0	0	1	16	127	237	125	7	0	0	0	(10)
(11)		CDD65	8	0	0	0	0	0	0	7	1	0	0	0	0	(11)
(12)		CDH74	163	0	0	0	0	1	31	113	17	1	0	0	0	(12)
(13)		CDH80	10	0	0	0	0	0	1	9	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.8	3.2	4.0	6.4	8.5	8.7	7.0	6.0	5.5	5.8	5.7	4.8	3.6	(14)
(15)	Precipitation	PrecAvg	5.9	0.0	0.1	0.2	0.2	0.4	1.1	2.0	1.2	0.5	0.2	0.1	0.1	(15)
(16)		PrecMax	10.9	0.2	0.4	2.3	0.9	1.1	5.4	5.7	3.6	1.5	1.0	0.3	0.2	(16)
(17)		PrecMin	2.3	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.0	0.0	0.1	0.4	0.2	0.4	1.0	1.4	0.7	0.4	0.2	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.8	31.0	47.0	61.2	72.0	77.8	81.6	76.2	68.8	55.7	42.9	26.2	(19)
(20)		MCWB	18.1	24.3	35.3	46.7	51.7	57.6	62.4	58.7	50.8	42.6	33.6	21.7	(20)	
(21)		2%	DB	17.4	25.5	41.4	55.7	66.9	73.7	77.3	72.9	64.0	50.5	36.1	21.1	(21)
(22)		MCWB	14.0	20.3	31.2	41.8	50.1	55.5	59.2	56.2	48.6	39.5	29.0	17.1	(22)	
(23)		5%	DB	12.7	21.5	36.8	51.6	62.6	70.6	73.9	69.8	60.3	46.5	31.1	17.0	(23)
(24)		MCWB	10.0	17.5	28.4	39.4	47.6	54.4	57.1	54.7	46.7	36.1	25.8	13.8	(24)	
(25)		10%	DB	8.3	16.9	31.8	47.5	58.7	67.5	70.7	66.8	56.7	42.3	26.1	12.5	(25)
(26)		MCWB	6.2	13.7	25.3	36.7	44.7	52.2	55.3	53.0	44.7	33.8	21.8	10.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	24.7	36.6	48.1	55.7	61.1	65.7	61.7	53.6	44.3	34.1	21.9	(27)
(28)		MDCB	21.9	30.4	45.6	58.3	69.0	75.1	78.7	73.9	65.1	54.5	42.0	25.9	(28)	
(29)		2%	WB	14.0	20.5	31.8	43.8	51.9	58.3	62.2	58.6	50.6	40.2	29.7	17.4	(29)
(30)		MDCB	17.3	25.0	39.9	53.8	64.0	70.8	73.8	70.6	61.3	49.3	36.0	20.8	(30)	
(31)		5%	WB	10.0	17.5	28.5	40.1	48.6	56.1	59.7	56.4	48.0	37.0	25.9	13.7	(31)
(32)		MDCB	12.7	21.3	36.2	50.1	60.6	68.0	71.1	66.9	58.5	45.3	31.2	16.8	(32)	
(33)		10%	WB	6.2	13.8	25.4	37.0	45.7	53.7	57.3	54.2	45.3	34.0	21.8	10.2	(33)
(34)		MDCB	8.3	16.8	31.8	46.5	56.9	65.0	68.0	63.9	55.0	41.8	25.9	12.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	26.8	29.2	27.9	23.7	23.7	22.9	21.9	23.3	26.3	26.2	25.5	24.5	(35)	
(36)		5% DB	MCDBR	30.3	32.8	31.6	29.7	31.1	28.9	27.7	28.9	32.1	32.4	30.8	27.5	(36)
(37)		MCWBR	27.2	28.2	23.6	19.9	19.4	17.0	14.9	16.1	20.2	23.0	24.7	24.0	(37)	
(38)		5% WB	MCDBR	30.0	32.6	30.8	28.8	28.8	26.6	25.2	26.0	30.6	31.5	30.2	27.0	(38)
(39)		MCWBR	27.0	28.5	23.8	21.2	20.2	17.6	16.1	16.5	21.1	23.6	24.5	23.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.216	0.244	0.285	0.324	0.334	0.344	0.342	0.319	0.286	0.263	0.236	0.218	(40)	
(41)		taud	2.402	2.336	2.254	2.235	2.261	2.320	2.366	2.409	2.456	2.443	2.414	2.387	(41)	
(42)		Ebn at Noon	292	300	299	295	295	291	290	294	297	289	279	276	(42)	
(43)		Edh at Noon	21	28	36	41	41	39	37	34	29	25	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	618	968	1448	1892	2204	2181	2073	1857	1577	1076	663	500	(44)	
(45)		RadStd	47	49	62	70	134	113	106	82	68	61	48	33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	-3.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-3.98	N/A	N/A	-2.57	N/A	N/A	N/A	N/A

Nomenclature: See separate page